

StateFi: Effectively Identifying Wi-Fi Devices through State Transitions

Abhishek K. Mishra^{2,1} , Mathieu Cunche¹

¹*Inria*, ²*Karlsruhe Institute of Technology (KIT)*

Context

- Unencrypted Wi-Fi management frames enable MAC based *tracking*
- *MAC randomization* breaks it
- Existing *probe-based signatures* (IEs, SEQ, timing, RSSI) are *brittle*
- Performance *drops* in real-world deployments

StateFi

- **StateFi:** behavioral fingerprinting via *full management-frame FSMs*
- Exploits full-frame transitions and timing dynamics
- **Behavioral Information Captured:**
 - *Protocol implementation logic (OS/vendor specific)*
 - *Retry & self-transition patterns*
 - *Scanning and connection behavior*
 - *Timing & power-saving strategies*

FSM-Based Device Characterization

Data Collection and Preprocessing

We use eight datasets:

- Campus datasets consist of all management frames
- Three widely used public probe datasets: **Infocom'21**, **Cagliari**, and **MITIK**

Env.	Duration (min)	MACs	Packets	Mgmt.
Residence	80	1,526	304k	51k
Cafe	68	1,782	80k	21k
Laboratory	66	2,448	321k	35k
Classroom	93	4,726	541k	59k
Restaurant	85	4,735	207k	43k
Total	–	31,862	2.6M+	235k

Summary of the Campus dataset

- *Burst is a sequence of packets emitted by the same MAC address where inter-packet gap $< 1s$*
- *Bursts with MAC are assigned a unique anonymized identifier, enabling ground-truth*

Modelling Device Behaviour as an FSM

$$F_d = (Q, \Sigma, \delta, q_0$$

)

- **States (Q):** Management frame subtypes (ProbeReq, Auth, AssocReq, ...)
- **Input (Σ):** Timestamp-ordered management frames
- **Transitions (δ):** Consecutive frame-type transitions
- **Initial state (q_0):** ProbeReq (probe-only) or first observed frame

(full FSM)

➡ **Passive traces** → **Directed FSM**: add edge $q_i \rightarrow q_j$ whenever frame j immediately follows frame i

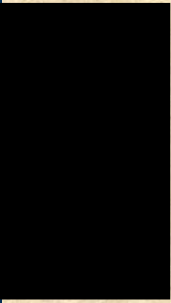
Constructing FSMs from Bursts

Two FSM Variants

Full-Management FSM	Probe-Only FSM
Stable MAC addresses	Randomized MAC addresses
Uses all management frames	Uses probe-phase frames only
Captures complete protocol behavior	Captures scanning behavior
In-network fingerprinting	Pre-association re-identification

Burst-Level Construction:

- Build one FSM **per management-frame burst**
- Parse ordered frame subtypes → construct a **directed, weighted transition graph**
- Consecutive identical frames create **self-transitions**
- Record both **transition frequencies** and **inter-state delays**



FSM-Based Fingerprinting

FSM Feature Representation

Each burst-level FSM is converted into a **feature vector** for classification

Feature	Description
x_1	Number of unique FSM states
x_2	Total state transitions
x_3	Self-transitions
x_4	Transition entropy
x_5	Transition rate
x_6	Mean intra-burst time gap
x_7	Maximum sequence-number gap

Hybrid Representation

- **FSM features (x_1 – x_7):** Capture behavioral dynamics
- **IE vector (x_8):** Binary encoding of a random Probe Request Information Elements

Final fingerprint = Behavioral dynamics (FSM) + Protocol semantics

Device-Pair Classification

Pairwise Similarity

Features

For two burst FSMs $\mathbf{x}_i$ and $\mathbf{x}_j$, compute:

- **L2 distance** (Euclidean)
- **Cosine distance**
- **L1 distance** (Manhattan)
- **IE similarity** (Information Element

- **Models:** i) **Random Forest (RF)** , ii) **Support Vector Machine (SVM)** , iii) **Logistic Regression (LR)**
- **70/30** train-test split

Training Data

- **Positive pairs:** Bursts from the **same persistent MAC**
- **Negative pairs:** Bursts from **different MACs**

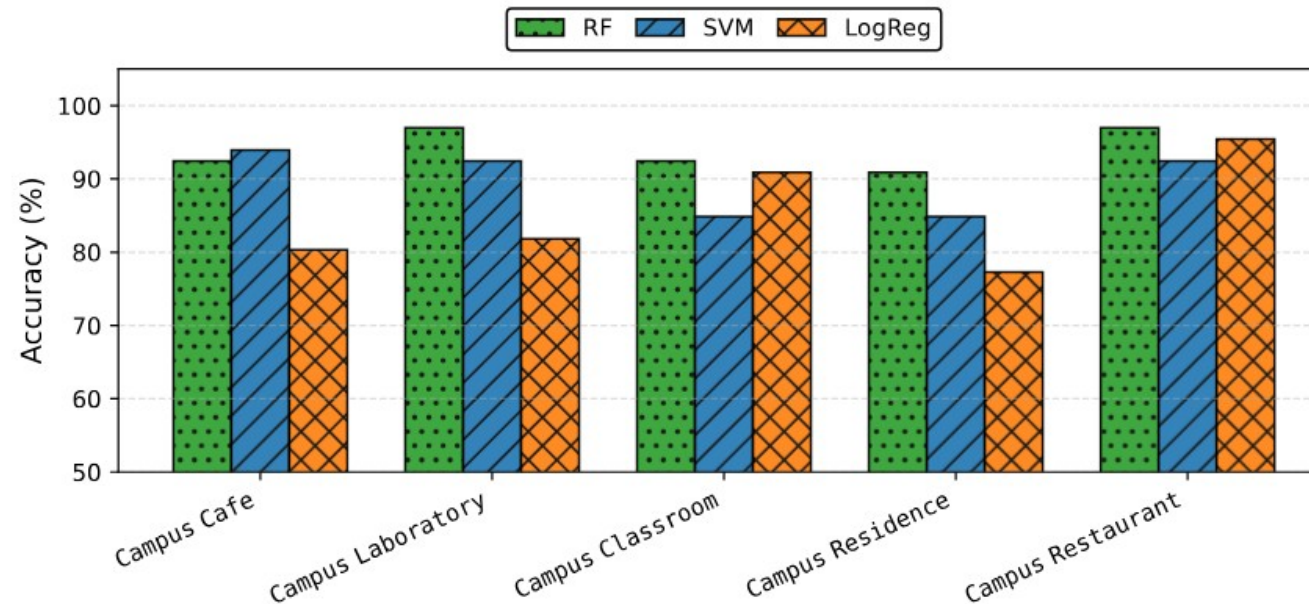
Balanced dataset ($\leq 5,000$ pairs per scenario)

Goal: Predict whether two bursts originate from the **same physical device** without directly comparing FSM graphs

Results

Accuracy In Fingerprinting devices

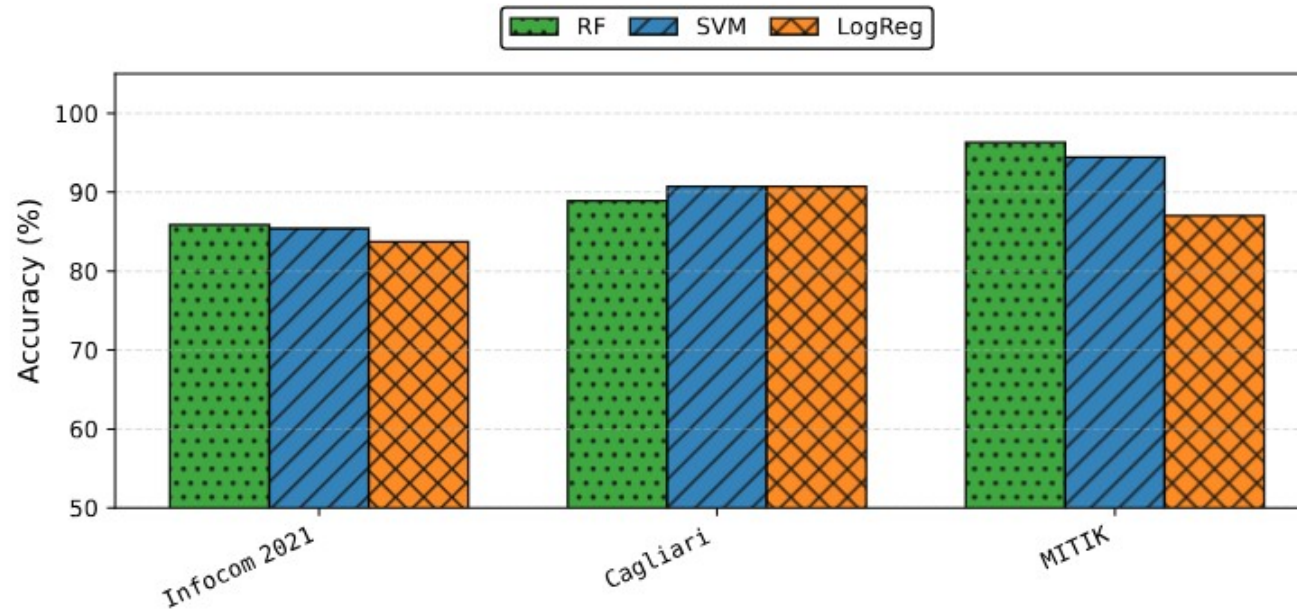
Goal is to fingerprint devices and detect anomalous behaviour even after association



RF yields the strongest performance, achieving approximately 94–97%

Accuracy in Defeating MAC Randomization

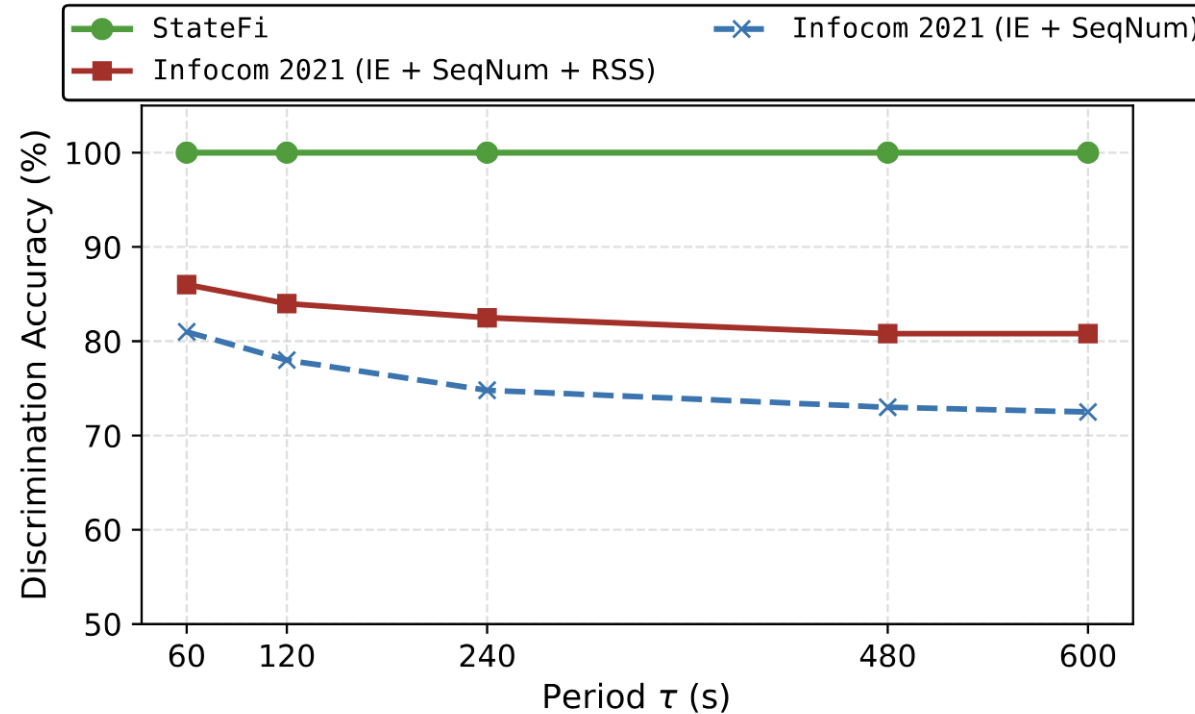
We restrict our FSMs to probe requests and responses and link bursts generated by the same physical device across randomized MACs



RF consistently achieves the highest accuracy (typically 92–97%)

Comparison with State-of-the-Art

We compare the discrimination accuracy with the SOTA framework: InfoCom 2021



StateFi achieves up to 98% discrimination accuracy and defeats SOTA

■ Outlook

Takeaways

- **StateFi:** First *FSM-based* fingerprinting of Wi-Fi devices
- Fingerprints device *behavior*, not *just packet contents*
- **94–97%** *in-network*, **92–97%** *probe-only* identification
- **Impact:** **12–17%** improvement *over SOTA* with single sniffer

■ Thank You! Questions?